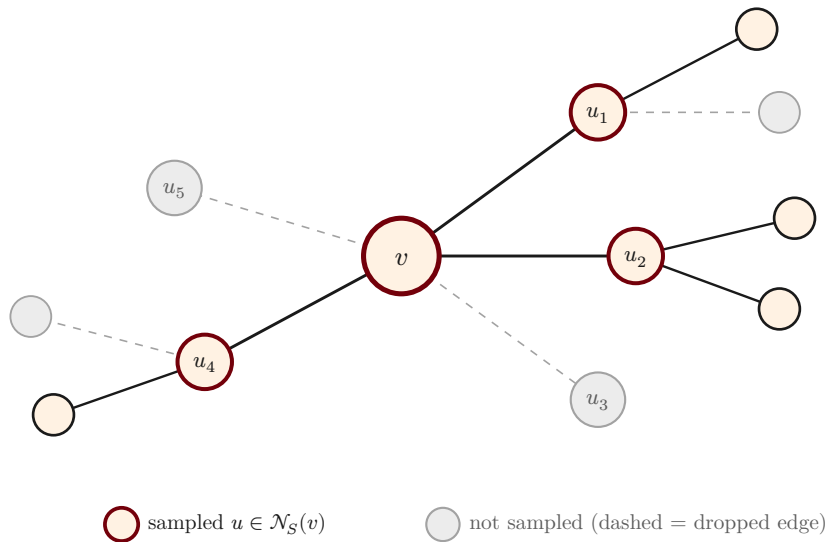


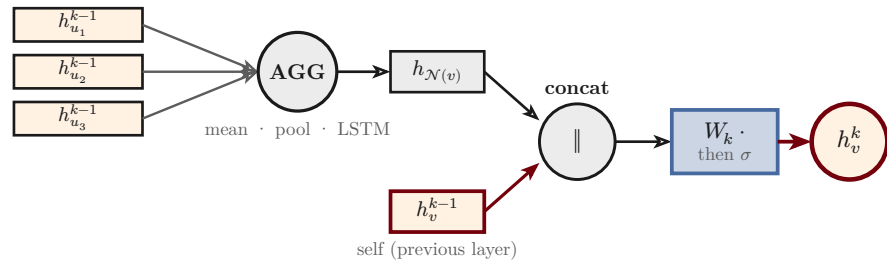
(a) fixed-size 2-hop neighbourhood sampling



hop-1: keep S_1 of $\deg(v)$ neighbours · hop-2: keep S_2 per node · bounded fan-out $S_1 S_2$

(b) layer- k update for node v

sampled neighbour features



$$h_v^k = \sigma(W_k \cdot [h_v^{k-1} \parallel \text{AGG}_k(\{h_u^{k-1} : u \in \mathcal{N}_S(v)\})])$$

repeat for $k = 1..K$ layers (K hops); optional ℓ_2 -normalise h_v^k